

>_ BEAST: The Inference Economy Inversion

Output Governance, Provider Fitness, and Compute Governance for Agentic Systems



[SYS_INIT] Working Paper v0.4 | Definitive Mega-Test Evidence Build Loaded... [OK]

[THE CURRENT ECONOMY]

- >_ Model calls = Rented repeatedly.
- >_ Cost = Tokens, latency, USD.
- >_ Outcome = Disposable service response.

>_

[THE BEAST INVERSION]

- >_ Verified work = Recorded local capability.
- >_ Cost = Measured against future avoidability.
- >_ Outcome = Durable local asset.

>_

[THE CORE THESIS]

Probabilistic compute is invoked for unresolved uncertainty, constrained into verifiable action, measured by provider fitness, recorded through compute receipts, and crystalized into reusable local capability.

[FALLACY: Cloud as Writer]

- >_ OUTPUT : Full file contents, massive diffs, formatting Python in JSON strings.
- >_ TOKENS : Exponential explosion. Rewrites code the local machine already has.
- >_ SAFETY : Unconstrained. Cloud model controls the execution.
- >_ RESULT : Brittle, expensive, slow.■

[SOLUTION: Cloud as Strategist]

- >_ OUTPUT : Context-light cloud intent (Action IR). Deliberately incomplete.
- >_ TOKENS : Tiny. Cheap. Points to local references (F2::BeastMissionConsole).
- >_ SAFETY : Cloud proposes high-level intent. BEAST governs local execution.
- >_ RESULT : Compact, verifiable, compiled locally.■

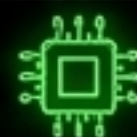
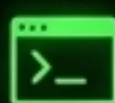
```
{
  'kind': 'action_intent.v1',
  'action': 'modify_symbol',
  'target': 'F2',
  'intent': 'ensure Enter submission is handled'
}
```



[Action IR Packet]

[BEAST Local Runtime]

>_ Resolve AST
/ Anchors



>_ Generate
Candidate Patch



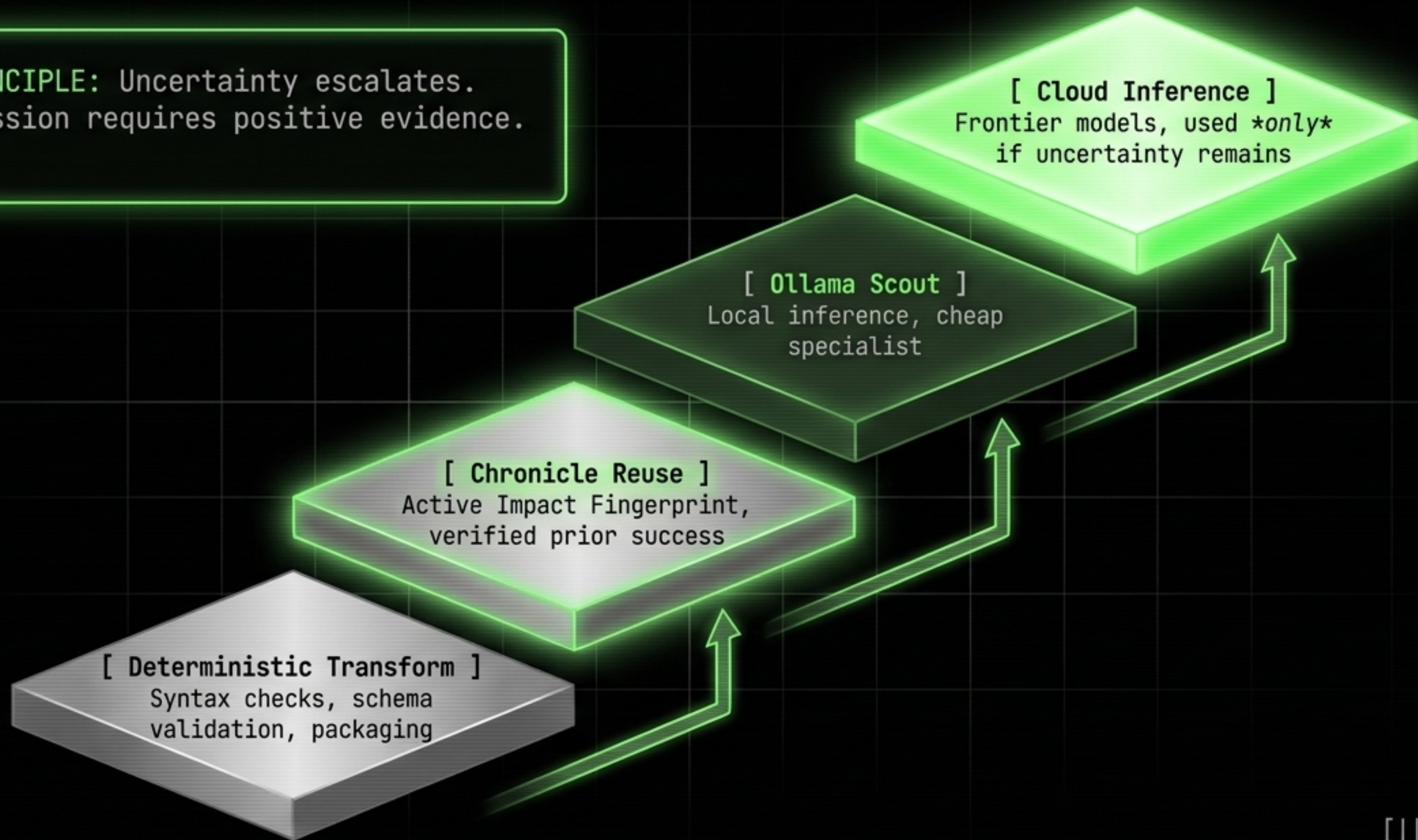
>_ Run PyTest
/ Verifier



[PAYOFF]

[**BEAST Action IR:** Context-light cloud intent, locally compiled into verified actions.]

>_ PRINCIPLE: Uncertainty escalates.
Suppression requires positive evidence.
█



[|||||.....]

[ComputePlan]

(Pre-Inference)

- >_ Budgets set:
cloud_calls: 1
input_tokens: 2500
- >_ Escalation ladder loaded.
- >_ Reuse candidates identified. ■

[ComputeGate]

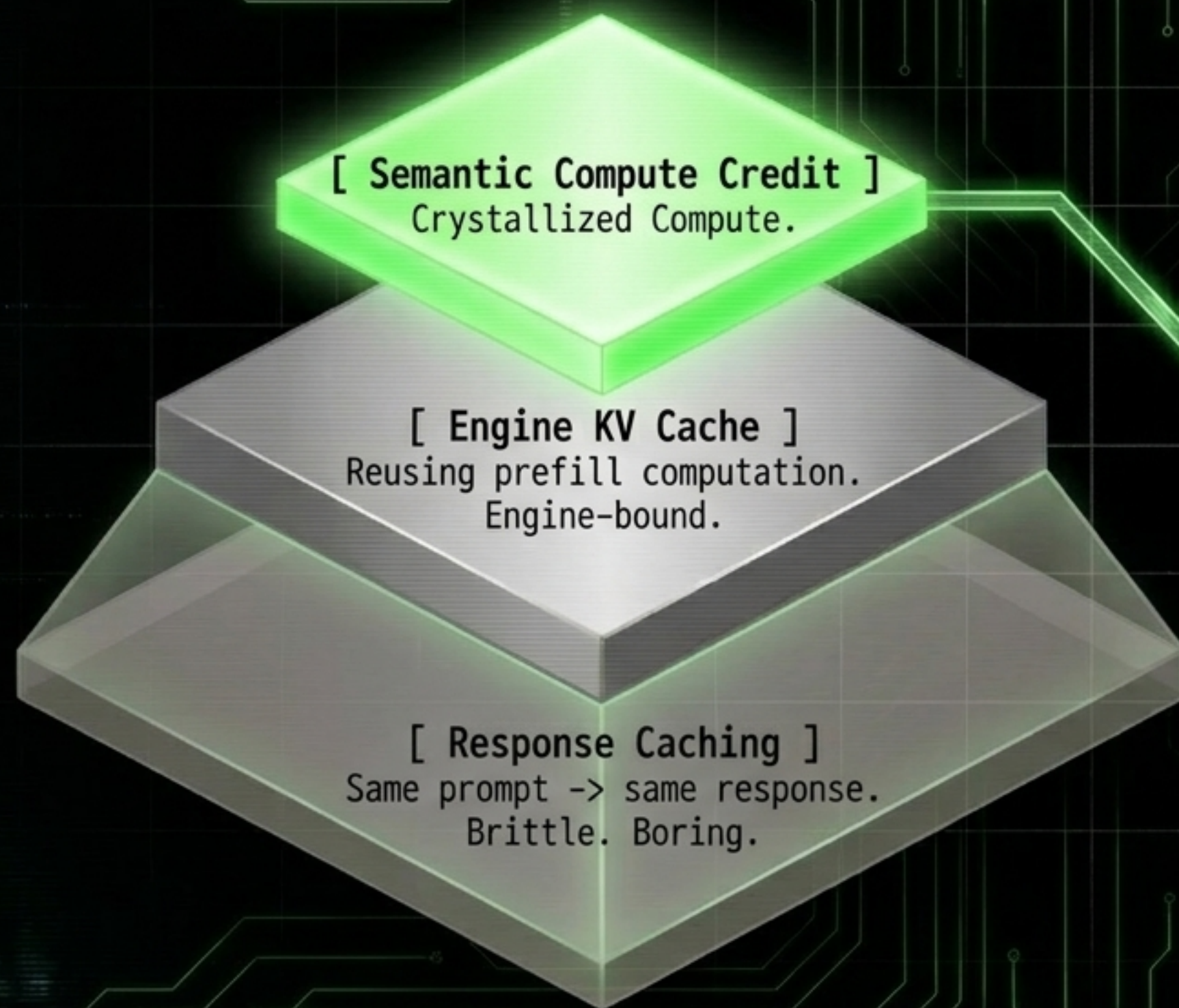
(Runtime Decision)

- >_ Threshold logic evaluation:
[reuse] vs [deterministic]
vs [escalate]
- >_ STATUS: Every escalation
must mathematically
mathematically justify why
probabilistic inference is
necessary. ■

[ComputeReceipt]

(Post-Inference)

- >_ Accounting record appended.
 - Work requested
 - Work avoided
 - Cost execution
 - Verification status
- >_ LESSON STATUS: Reusable.



[SEMANTIC COMPUTE CREDIT]

>_ WHAT IS IT?

A verified reusable artifact that reduces future uncertainty.

>_ CONSTITUENTS:

- Impact Fingerprint
- Verified Tests
- Avoided Tokens Estimate

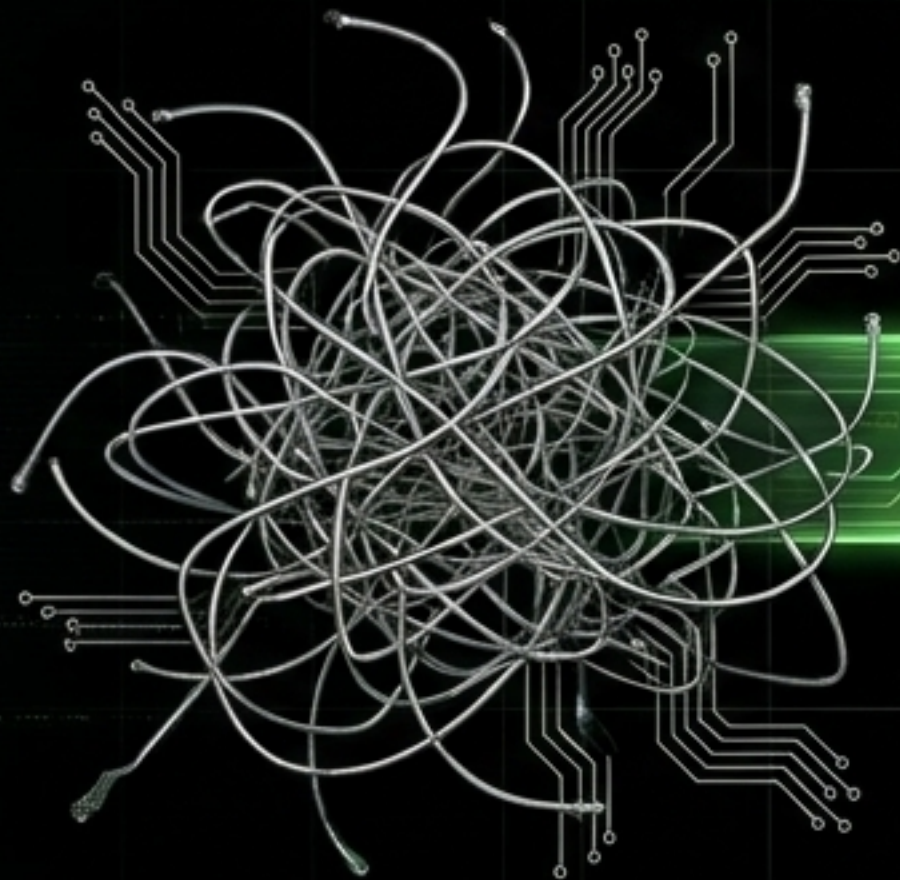
>_ WHY IT MATTERS:

It survives repository changes because it trusts **verified behavior under a fingerprint**, not just raw text.

[...:]
>_MISSION CONSOLE // THE EDGE GRINDER PROCESS VIEW]_

[The Compute Forge Node]

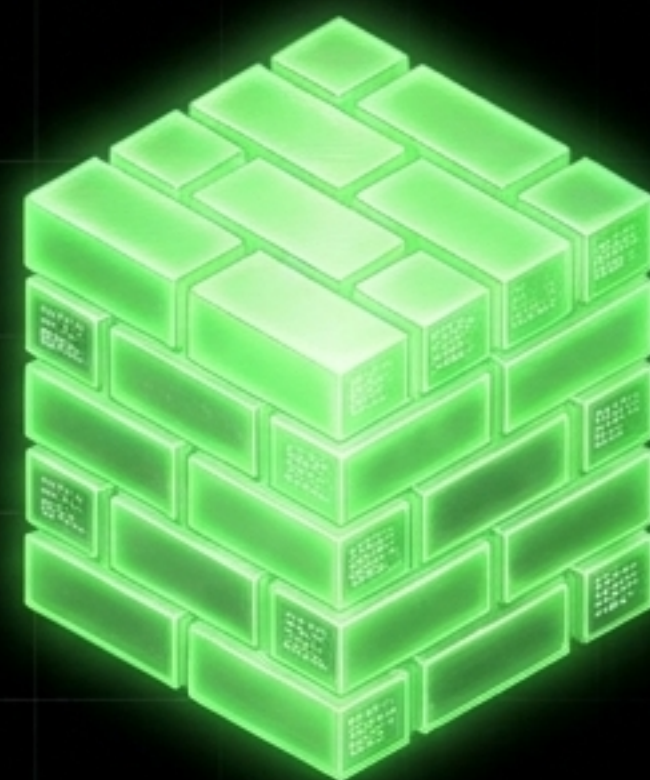
[Inference-Ready Flour]



[Raw Repository Material]



Not generating tokens.
Displacing probabilistic work.



>_ - AST maps
>_ - test-impact maps
>_ - deterministic patches
>_ - semantic chunks

>_ The Forge Node earns internal BEAST compute credits by displacing future cloud tokens.

[|||||.....]

[LANE C: FULL BEAST GOVERNANCE]

Completion Rate:

```
[|||||||||||||||||] 100% (30/30)
```

Status: Fully verified, compiled, and Chronicle-recorded.

>_

business's governance/rescue path
output on these coding fixtures.

[MATURE NVIDIA CRYSTALLIZATION TRAJECTORY]



[QPCCD RESULT METRICS]

>_ Mature NVIDIA Run: 72 controlled observations.

>_ QPCCD Result: 12 deterministic zero-call reuses out of 24 mature observations (0.50 QPCCD).

MEANING:
Capability crystallizes entirely at Occurrence 3.
Cloud calls drop to absolute zero.

// MUTATION SAFETY INITIATED: Occurrence-10 Drift

>_ [BLOCKED] 6/6 stale reuses
blocked during material drift.

>_ [RECOVERED] 6/6 recovered
through fresh verification.

>_ [FALSE REUSES] 0 recorded.

[The A-D Escalation Ladder]

>_ A: Cosmetic drift
(Remains active).

>_ B: Semantic drift
(Shadow revalidation).

>_ C: Schema drift
(Shadow revalidation).

>_ D: Breaking target drift
(Demoted. Requires
cloud/human escalation). █

BEAST Capability Treasury: Provider-Independent Transfer Flow

> _ TRANSFER METRICS █

> _ Target-Provider Calls: 0
> _ Successful Transfers: 12 / 12
> _ Estimated Avoided Tokens: 18,768



Prove provider-independent **consumption** of a verified local capability.
The causal object is the repository-bound fingerprint, not the cloud model.

[The Semantic Work Queue]

[Cloud
Inference]

(Invoked only for
uncertainty)

[Action IR
& Verifier]

(Locally compiled
and proven)

[Impact Fingerprint
& Stored Artifact]

(Crystallized into the
Capability Treasury)

The Compute Governor

(Routes the next task to the
Forge instead of the Cloud)

[Compute
Forge Node]

(Grinds artifacts
into deterministic
transforms)

> **INFERENCE COMPOSTING:** Yesterday's probabilistic model work
becomes tomorrow's deterministic soil.

**"BEAST does not merely automate work.
It standardizes the meaning of work."**

- >_ Every task becomes memory.
- >_ Every memory becomes leverage.
- >_ Every repeated success becomes infrastructure.